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Surgical instrument and method

Abstract

A surgical instrument includes a first member being engageable to an inner surface of a spinal implant. An actuator is connected to a second member such that the second member is translatable relative to the first member for connecting the spinal implant to a bone fixation device. Systems, spinal constructs, implants and methods are disclosed.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/078,648, filed Oct. 23, 2020, which is expressly incorporated by reference herein, in its entirety.

TECHNICAL FIELD

(1) The present disclosure generally relates to medical devices for the treatment of musculoskeletal disorders, and more particularly to a surgical system and a method for treating a spine.

BACKGROUND

(2) Spinal disorders such as degenerative disc disease, disc herniation, osteoporosis, spondylolisthesis, stenosis, scoliosis, kyphosis and other curvature abnormalities, tumor, and fracture may result from factors including trauma, disease and degenerative conditions caused by injury and aging. Spinal disorders typically result in symptoms including pain, nerve damage, and partial or complete loss of mobility.

(3) Non-surgical treatments, such as medication, rehabilitation and exercise can be effective, however, may fail to relieve the symptoms associated with these disorders. Surgical treatment of these spinal disorders includes correction, fusion, fixation, discectomy, laminectomy and implantable prosthetics. As part of these surgical treatments, spinal constructs, which include implants such as bone fasteners, connectors, plates and vertebral rods are often used to provide stability to a treated region. These implants can redirect stresses away from a damaged or defective region while healing takes place to restore proper alignment and generally support the vertebral members. Surgical instruments are employed, for example, to engage the fasteners for attachment of implants to the exterior of two or more vertebral members. This disclosure describes an improvement over these prior technologies.

SUMMARY

(4) In one embodiment, a surgical instrument is provided. The surgical instrument includes a first member being engageable to an inner surface of a spinal implant. An actuator is connected to a second member such that the second member is translatable relative to the first member for connecting the spinal implant to a bone fixation device. In some embodiments, systems, spinal constructs, implants and methods are disclosed.

(5) In one embodiment, the surgical instrument includes an outer sleeve including a distal portion having at least one spring tab biased radially inward and expandable for engagement to an inner surface of a spinal implant. An inner shaft includes a distal portion and a handle is connected to the inner shaft such that the distal portion of the inner shaft is translatable relative to the distal portion of the outer sleeve for connecting the spinal implant to a bone fastener shaft.

(6) In one embodiment, a surgical system is provided. The surgical system includes a bone fastener shaft configured for fixation to vertebral tissue. A spinal implant receiver is configured for connection to the bone fastener shaft. A surgical instrument includes a first member that is engageable to an inner surface of the spinal implant receiver and a second member. The surgical instrument further includes an actuator connected to the second member such that the second member is translatable relative to the first member for connecting the spinal implant receiver to the bone fastener shaft.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure will become more readily apparent from the specific description accompanied by the following drawings, in which:

(2) FIG. 1 is a perspective view of components of one embodiment of a surgical system in accordance with the principles of the present disclosure;

(3) FIG. 2 is a side cross section view of the components shown in FIG. 1;

(4) FIG. 3 is a break away view of components of one embodiment of a surgical system in accordance with the principles of the present disclosure;

(5) FIG. 4 is a break away cross section view of components of one embodiment of a surgical system in accordance with the principles of the present disclosure;

(6) FIG. 5 is a perspective view of the components shown in FIG. 1;

(7) FIG. 6 is a side cross section view of the components shown in FIG. 1;

(8) FIG. 7 is a break away view of components of one embodiment of a surgical system in accordance with the principles of the present disclosure;

(9) FIG. 8 is a cross section view of the components shown in FIG. 7;

(10) FIG. 9 is a side cross section view of the components shown in FIG. 1;

(11) FIG. 10 is a perspective view of the components shown in FIG. 1;

(12) FIG. 11 is a cross section view of the components shown in FIG. 10;

(13) FIG. 12 is a break away view of components of one embodiment of a surgical system in accordance with the principles of the present disclosure;

(14) FIG. 13 is a cross section view of the components shown in FIG. 12;

(15) FIG. 14 is a perspective view of components of one embodiment of a surgical system in accordance with the principles of the present disclosure;

(16) FIG. 15 is a perspective view of components of one embodiment of a surgical system in accordance with the principles of the present disclosure;

(17) FIG. 16 is a perspective view of the components shown in FIG. 15;

(18) FIG. 17 is a break away view of components of one embodiment of a surgical system in accordance with the principles of the present disclosure;

- (19) FIG. 18 is a cross section view of the components shown in FIG. 17;
- (20) FIG. 19 is an enlarged break away view of the components shown in FIG. 15;
- (21) FIG. 20 is a cross section view of the components shown in FIG. 17;
- (22) FIG. 21 is a cross section view of the components shown in FIG. 17; and
- (23) FIG. 22 is a cross section view of the components shown in FIG. 17.

DETAILED DESCRIPTION

(24) The exemplary embodiments of a surgical system are discussed in terms of medical devices for the treatment of musculoskeletal disorders and more particularly, in terms of a surgical system and a method for treating a spine. In some embodiments, the present surgical system includes a surgical instrument, for example, an inserter engageable to a spinal implant, for example, a receiver. In some embodiments, the present surgical system includes an inserter having an end engageable to an inner surface of a spinal implant to capture and connect the spinal implant to a bone fixation device, including a bone fastener shaft.

(25) In some embodiments, the present surgical system includes a surgical instrument, for example, an inserter configured to connect a spinal implant to a bone fixation device, including a bone fastener shaft. In some embodiments, the inserter includes an outer sleeve, an inner shaft and an actuator. In some embodiments, the actuator is connected to the inner shaft such that the inner shaft is translatable relative to the outer sleeve for connecting the spinal implant to the bone fastener shaft. In some embodiments, the inserter is configured to connect a spinal implant to bone fixation devices, including bone screws, hooks and/or plates.

(26) In some embodiments, the present surgical system includes a surgical instrument, for example, an inserter configured for utilization with a modular screw platform. In some embodiments, the modular screw platform includes a plurality of spinal implants, for example, a plurality of receivers having a plurality of heights and/or a plurality of external geometries. In some embodiments, an end of the inserter is configured to engage to the plurality of receivers. In some embodiments, the end of the inserter engages rocker holes of the receiver via an inner surface of the receiver to secure the receiver to the end of the inserter.

(27) In some embodiments, the present surgical system includes a surgical instrument, for example, an inserter having an outer sleeve and an inner shaft with a low profile configuration. In some embodiments, the low profile configuration includes an end of the inserter configured to fit within a profile of a spinal implant, for example, a receiver to secure the receiver to the end of the inserter. In some embodiments, a profile of the spinal implant includes an interior surface that defines an internal groove. In some embodiments, the internal groove is disposed proximal to a receiver crown. In some embodiments, the outer sleeve engages the internal groove to secure the inserter to the receiver.

(28) In some embodiments, the present surgical system includes a surgical inserter configured for securement to a spinal implant receiver. In some embodiments, the inserter includes an outer sleeve, an inner shaft and an actuator. In some embodiments, the outer sleeve includes an end including spring tabs that are biased radially inward and are expandable to engage to an inner surface of the receiver. In some embodiments, to secure the inserter to the receiver, the receiver is loaded to the distal end of the inserter. In some embodiments, the spring tabs are bent inwardly to enable the inserter to pass through the receiver. In some embodiments, the spring tabs are bent inwardly to enable the inserter to pass through a minor diameter of a thread of the inner surface of the receiver. In some embodiments, the spring tabs are heat treated in a bent position to bias the spring tabs inwardly into a closed position. In some embodiments, the receiver is secured to the inserter. In some embodiments, the inner shaft is translated toward the distal end of the inserter to deploy the spring tabs in an outward direction. In some embodiments, deployment of the spring tabs in the outward direction engages the spring tabs to the inner surface of the receiver to secure the receiver to the inserter. In some embodiments, the receiver is locked to the inserter. In some embodiments, the inner shaft is translatable through the spring tabs to push a crown of the spinal

implant in a downward direction to engage a head of a screw shaft.

(29) In some embodiments, the present surgical system includes a surgical inserter configured for securement to a spinal implant receiver. In some embodiments, the inserter includes a first member and a second member. In some embodiments, the first member includes an outer sleeve and the second member includes an inner shaft. In some embodiments, the inserter includes an actuator, for example, a pair of handles. In some embodiments, the actuator is connected to the inner shaft such that the inner shaft is translatable relative to the outer sleeve for securing the receiver to a bone fixation device, including a bone fastener shaft. In some embodiments, the inserter includes a latch, for example, a locking mechanism including a finger engagement surface. In some embodiments, the latch is connected to the actuator in at least one non-locked orientation such that the actuator is movable relative to the inner shaft and a locked orientation such that the actuator is fixed relative to the inner shaft. In some embodiments, the actuator is movable between positions such that the inserter can be secured and unsecured to the inserter. In some embodiments, the actuator is movable between an open position including the non-locked orientation, an intermediate position including the locked orientation and a closed position including the non-locked orientation. In some embodiments, the actuator translates the inner shaft relative to the outer sleeve via a threaded engagement.

(30) In some embodiments, the actuator is oriented in the open position. In some embodiments, in the open position, the receiver is loaded into the distal end. In some embodiments, in the open position, the receiver can be loaded and/or removed from the distal end of the inserter. In some embodiments, in the open position, spring tabs of the outer sleeve are in a flexed inward state such that the spring tabs can be inserted into the interior features of the receiver. In some embodiments, in the open position, the spring tabs are in a collapsed inward state to slide through a minor diameter of an interior of the receiver.

(31) In some embodiments, the actuator is oriented in the intermediate position. In some embodiments, in the intermediate position, the actuator is locked until the latch is manually depressed by a user. In some embodiments, the latch locks onto a pin that is centrally disposed on a shaft of the member to prevent the actuator from opening or closing. In some embodiments, in the intermediate position, the receiver is secured to the end of the inserter. In some embodiments, in the intermediate position, the outer sleeve remains fixed and the inner shaft translates in a direction, for example, axially. In some embodiments, the inner shaft translates a distance of, for example 2.0 to 3.0 mm. In some embodiments, the actuator is released to disconnect the receiver from the distal end. In some embodiments, the spring tabs are forced in an outward orientation by translation of the inner shaft to engage to the interior features of the receiver.

(32) In some embodiments, the actuator is oriented in the closed position. In some embodiments, the latch is depressed and the actuator is compressed to position the inserter in the locking orientation to lock the receiver to the distal end. In some embodiments, in the closed position, the actuator is released when the actuator is compressed. In some embodiments, in the closed position, the outer sleeve remains fixed and the inner shaft translates in a direction, for example, axially. In some embodiments, the inner shaft translates a distance of 6.0 mm. In some embodiments, in the closed position, the receiver is secured to the end of the inserter via the spring tabs that remain in the outward orientation as the inner shaft translates axially beyond the spring tabs. In some embodiments, a tip of the inner shaft pushes an implant crown in a downward direction and into the receiver.

(33) In some embodiments, the present surgical system includes a surgical inserter employed with a method for connecting a spinal implant receiver to a bone fixation device, including a bone fastener shaft. In some embodiments, the method includes the step of introducing an inserter. In some embodiments, the inserter includes a proximal end and a distal end. In some embodiments, the inserter includes an outer sleeve, an inner shaft, a latch and an actuator. In some embodiments, the method includes the step of disposing the inserter in an initial open position to load an implant, for

example, a receiver to the distal end. In some embodiments, in the open position, the receiver is loaded into the distal end. In some embodiments, the method includes the step of disposing the inserter in an intermediate position to secure the receiver to the distal end. In some embodiments, in the intermediate position, the actuator is locked until the latch is manually depressed by a user. In some embodiments, the method includes the step of disposing the inserter into a closed position to lock the receiver to the distal end. In some embodiments, in the closed position, the actuator is released when the actuator is compressed.

(34) In some embodiments, the system of the present disclosure may be employed to treat spinal disorders, for example, degenerative disc disease, disc herniation, osteoporosis, spondylolisthesis, stenosis, scoliosis and other curvature abnormalities, kyphosis, tumor and fractures. In some embodiments, the system of the present disclosure may be employed with other ostial and bone related applications, including those associated with diagnostics and therapeutics. In some embodiments, the disclosed system may be alternatively employed in a surgical treatment with a patient in a prone or supine position, and/or employ various surgical approaches to the spine, including anterior, posterior, posterior mid-line, direct lateral, postero-lateral, and/or antero-lateral approaches, and in other body regions. The system of the present disclosure may also be alternatively employed with procedures for treating the lumbar, cervical, thoracic, sacral and pelvic regions of a spinal column. The system of the present disclosure may also be used on animals, bone models and other non-living substrates, for example, in training, testing and demonstration.

(35) The system of the present disclosure may be understood more readily by reference to the following detailed description of the embodiments taken in connection with the accompanying drawing figures, which form a part of this disclosure. It is to be understood that this application is not limited to the specific devices, methods, conditions or parameters described and/or shown herein, and that the terminology used herein is for the purpose of describing particular embodiments by way of example only and is not intended to be limiting. In some embodiments, as used in the specification and including the appended claims, the singular forms “a,” “an,” and “the” include the plural, and reference to a particular numerical value includes at least that particular value, unless the context clearly dictates otherwise. Ranges may be expressed herein as from “about” or “approximately” one particular value and/or to “about” or “approximately” another particular value. When such a range is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another embodiment. It is also understood that all spatial references, for example, horizontal, vertical, top, upper, lower, bottom, left and right, are for illustrative purposes only and can be varied within the scope of the disclosure. For example, the references “upper” and “lower” are relative and used only in the context to the other, and are not necessarily “superior” and “inferior”.

(36) As used in the specification and including the appended claims, “treating” or “treatment” of a disease or condition refers to performing a procedure that may include administering one or more drugs to a patient (human, normal or otherwise or other mammal), employing implantable devices, and/or employing instruments that treat the disease, for example, microdiscectomy instruments used to remove portions bulging or herniated discs and/or bone spurs, in an effort to alleviate signs or symptoms of the disease or condition. Alleviation can occur prior to signs or symptoms of the disease or condition appearing, as well as after their appearance. Thus, treating or treatment includes preventing or prevention of disease or undesirable condition (e.g., preventing the disease from occurring in a patient, who may be predisposed to the disease but has not yet been diagnosed as having it). In addition, treating or treatment does not require complete alleviation of signs or symptoms, does not require a cure, and specifically includes procedures that have only a marginal effect on the patient. Treatment can include inhibiting the disease, e.g., arresting its development, or relieving the disease, e.g., causing regression of the disease. For example, treatment can include reducing acute or chronic inflammation; alleviating pain and mitigating and inducing re-growth of

new ligament, bone and other tissues; as an adjunct in surgery; and/or any repair procedure. Also, as used in the specification and including the appended claims, the term “tissue” includes soft tissue, ligaments, tendons, cartilage and/or bone unless specifically referred to otherwise.

(37) The following discussion includes a description of a surgical system including a surgical instrument, related components and methods of employing the surgical system in accordance with the principles of the present disclosure. Alternate embodiments are also disclosed. Reference is made in detail to the exemplary embodiments of the present disclosure, which are illustrated in the accompanying figures. Turning to FIGS. **1-13**, there are illustrated components of a surgical system **10**.

(38) The components of surgical system **10** can be fabricated from biologically acceptable materials suitable for medical applications, including metals, synthetic polymers, ceramics and bone material and/or their composites. For example, the components of surgical system **10**, individually or collectively, can be fabricated from materials such as stainless steel alloys, aluminum, commercially pure titanium, titanium alloys, Grade 5 titanium, super-elastic titanium alloys, cobalt-chrome alloys, superelastic metallic alloys (e.g., Nitinol, super elasto-plastic metals, such as GUM METAL®), ceramics and composites thereof such as calcium phosphate (e.g., SKELITE™), thermoplastics such as polyaryletherketone (PAEK) including polyetheretherketone (PEEK), polyetherketoneketone (PEKK) and polyetherketone (PEK), carbon-PEEK composites, PEEK-BaSO₄ polymeric rubbers, polyethylene terephthalate (PET), fabric, silicone, polyurethane, silicone-polyurethane copolymers, polymeric rubbers, polyolefin rubbers, hydrogels, semi-rigid and rigid materials, elastomers, rubbers, thermoplastic elastomers, thermoset elastomers, elastomeric composites, rigid polymers including polyphenylene, polyamide, polyimide, polyetherimide, polyethylene, epoxy, bone material including autograft, allograft, xenograft or transgenic cortical and/or corticocancellous bone, and tissue growth or differentiation factors, partially resorbable materials, for example, composites of metals and calcium-based ceramics, composites of PEEK and calcium based ceramics, composites of PEEK with resorbable polymers, totally resorbable materials, for example, calcium based ceramics such as calcium phosphate, tri-calcium phosphate (TCP), hydroxyapatite (HA)-TCP, calcium sulfate, or other resorbable polymers such as polylactide, polyglycolide, polytyrosine carbonate, polycaprolactone and their combinations.

(39) The components of surgical system **10**, individually or collectively, may also be fabricated from a heterogeneous material such as a combination of two or more of the above-described materials. The components of surgical system **10** may be monolithically formed, integrally connected or include fastening elements and/or instruments, as described herein.

(40) Surgical system **10** can be employed, for example, with a minimally invasive procedure, including percutaneous techniques, mini-open and open surgical techniques to deliver and introduce instrumentation and/or components of spinal constructs at a surgical site within a body of a patient, for example, a section of a spine. In some embodiments, one or more of the components of surgical system **10** are configured for engagement with existing spinal constructs, which may include spinal implants such as one or more rods, fasteners, plates and connectors. In some embodiments, the spinal constructs can be attached with vertebrae in a revision surgery to manipulate tissue and/or correct a spinal disorder, as described herein.

(41) Surgical system **10** includes a surgical instrument, for example inserter **12**. Inserter **12** is configured for engagement to a spinal implant, for example, a receiver **14** of a bone fastener **16**, as shown in FIG. **3**. Inserter **12** is configured to secure receiver **14** to a bone fixation device, including shaft **18** of bone fastener **16** that has been implanted into a surgical site, for example, vertebral tissue, as described herein. Inserter **12** is configured for utilization with a modular screw platform such that inserter **12** can be implemented with various embodiments of receiver **14**.

(42) Inserter **12** includes a proximal end **20** and a distal end **22**, as shown in FIG. **1**. Inserter **12** extends along and defines a longitudinal axis X1, as shown in FIG. **2**. In some embodiments,

insertor **12** may have various cross-section configurations, for example, oval, oblong, triangular, rectangular, square, polygonal, irregular, uniform, non-uniform, variable, tubular and/or tapered. (43) Insertor **12** includes a member having an outer sleeve **26** and a member having an inner shaft **30**, as shown in FIGS. **3** and **4**. Sleeve **26** and shaft **30** are configured for engagement to receiver **14**. Sleeve **26** is configured for engagement to an inner surface of receiver **14**, as described herein. Sleeve **26** includes an end **34**, as shown in FIG. **5**. In some embodiments, sleeve **26** may have various cross-section configurations, for example, oval, oblong, triangular, rectangular, square, polygonal, irregular, uniform, non-uniform, variable, tubular and/or tapered.

(44) An end **32** is configured for engagement to an actuator **36**, as shown in FIG. **1**. An outer shaft **38** is disposed at end **32** and is configured for engagement to actuator **36** and a latch **40**, as shown in FIG. **2** and described herein. Latch **40** is connected to actuator **36** such that actuator **36** is movable relative to sleeve **26**. See also, for example, the embodiments and disclosure of an insertor and method for surgically treating a spine, shown and described in commonly owned and assigned U.S. patent application Ser. No. 17/078,631 filed Oct. 26, 2020, and published as U.S. Patent Application Publication No. 20220125487, on Apr. 28, 2022, the entire contents of which being incorporated herein by reference. In some embodiments, shaft **38** may have various cross-section configurations, for example, oval, oblong, triangular, rectangular, square, polygonal, irregular, uniform, non-uniform, variable, tubular and/or tapered.

(45) End **34** is configured for engagement to an inner surface of receiver **14**, as shown in FIGS. **3**, **4**, **7**, **8**, **12** and **13**. End **34** is expandable to engage the inner surface of receiver **14** to secure receiver **14** to insertor **12**. End **34** includes a spring tab **42** and a spring tab **44**. Tab **42** opposes tab **44**. Tabs **42** and **44** are flexible, biased radially inward and are expandable to engage the inner surface of receiver **14** and to secure receiver **14** to insertor **12**. Tab **42** includes an outer surface that defines a projection **46** and tab **44** includes an outer surface that defines a projection **48**, as shown in FIG. **4**. The inner surface of receiver **14** defines a groove **50** configured for engagement to projections **46** and **48**, as shown in FIG. **4**. In some embodiments, sleeve **26** includes one or more spring tabs.

(46) Shaft **30** includes an end **52** and an end **54**, as shown in FIG. **2**. Shaft **30** is in co-axial alignment relative to sleeve **26** and extends along longitudinal axis **X1**, as shown in FIG. **2**. Shaft **30** translates relative to sleeve **26** to secure receiver **14** to shaft **18**, as shown in FIGS. **3**, **4**, **7**, **8**, **12** and **13**. Shaft **30** is configured to expand sleeve **26** into engagement to the inner surface of receiver **14** to secure receiver **14**. In some embodiments, shaft **30** may have various cross-section configurations, for example, oval, oblong, triangular, rectangular, square, polygonal, irregular, uniform, non-uniform, variable, tubular and/or tapered.

(47) Sleeve **26** is configured for engagement to an end **56** of shaft **38**, as shown in FIG. **6**. Sleeve **26** includes a threaded portion **58** that is configured for disposal into a threaded recess **60** of end **56**, as shown in FIG. **6**. Sleeve **26** is fixed to shaft **38**. Shaft **30** is fixed with end **32** via a pin **33** and movable relative to sleeve **26**, as shown in FIGS. **1**, **2**, **6**, **9** and **11**. A biasing member, for example, a spring **62** is configured for disposal about end **56**, as shown in FIG. **9** and is configured to provide energy in an axial direction to facilitate movement of shaft **30** when actuator **36** is released, as described herein.

(48) End **54** is configured for engagement to a crown **64** disposed within a cavity **66**, as shown in FIG. **4**, of receiver **14**, as shown in FIGS. **3**, **7** and **12**. Crown **64** is configured for locking receiver **14** to shaft **18** in a snap fit assembly, as described herein. In some embodiments, end **54** may have various cross-section configurations, for example, oval, oblong, triangular, rectangular, square, polygonal, irregular, uniform, non-uniform, variable, tubular and/or tapered.

(49) End **20** of insertor **12** includes actuator **36**, as shown in FIG. **1**. Actuator **36** is connected to shaft **30** such that shaft **30** translates relative to sleeve **26** for connecting receiver **14** to shaft **18**. Actuator **36** is rotatable relative to sleeve **26** and is disposable between an open position including a non-locked orientation such that actuator **36** is movable relative to sleeve **26** (FIGS. **1** and **2**), an

intermediate position including a locked orientation such that actuator **36** is fixed relative to sleeve **26** (FIGS. **5** and **6**) and a closed position including a non-locked orientation such that actuator **36** is movable relative to sleeve **26** (FIGS. **10** and **11**), as described herein. In a natural state, actuator **36** is biased to the open position and is automatically movable from the closed position to the open position.

(50) Actuator **36** includes a pair of lever handles **68**, **70**, as shown in FIG. **2** that are rotatable relative to sleeve **26**. In some embodiments, handles **68**, **70** may have various cross-section configurations, for example, oval, oblong, triangular, rectangular, square, polygonal, irregular, uniform, non-uniform, variable, tubular and/or tapered. In some embodiments, an outer surface of handles **68**, **70** has one or more of various surface configurations, for example, rough, threaded, arcuate, undulating, porous, semi-porous, dimpled, polished and/or textured. In some embodiments, actuator **36** includes one or more handles.

(51) Handle **68** includes a bar linkage **72** that is rotatably engaged to end **32**, as shown in FIG. **2**. Linkage **72** includes an end **74** and an end **76**. End **74** includes a surface that defines an opening **78**. A surface of handle **68** defines an opening **80**. End **74** engages handle **68** via a pin **82** that is disposed within openings **78** and **80**. End **76** includes a surface that defines an opening **84**. A surface of end **32** defines an opening **86**. End **76** engages end **32** via a pin **88** that is disposed within openings **84** and **86**. Engagement between end **74** of linkage **72** and handle **68** creates a pivot point **P1**, as shown in FIG. **6**. Engagement between end **76** of linkage **72** and end **32** creates a pivot point **P2**, as shown in FIG. **6**.

(52) Handle **70** includes a bar linkage **90** rotatably engaged to end **32**, as shown in FIG. **2**. Linkage **90** includes an end **92** and an end **94**. End **92** includes a surface that defines an opening **96**. A surface of handle **70** defines an opening **98**. End **92** engages handle **70** via a pin **100** that is disposed within openings **96** and **98**. End **94** includes a surface that defines an opening **102**. A surface of end **32** defines an opening **104**. End **94** engages end **32** via a pin **106** that is disposed within openings **102** and **104**. Engagement between end **92** of linkage **90** and handle **70** creates a pivot point **P3**, as shown in FIG. **6**. Engagement between end **94** of linkage **90** and end **32** creates a pivot point **P4**, as shown in FIG. **6**.

(53) Latch **40** is connected to actuator **36** in a non-locked orientation (FIGS. **1-2** and **10-11**) such that actuator **36** is movable relative to sleeve **26** in the open position and the closed position, and a locked orientation (FIGS. **5** and **6**) such that actuator **36** is fixed relative to sleeve **26** in the intermediate position. Latch **40** is connected to actuator **36** via handle **68**, as shown in FIG. **2**. A surface of handle **68** defines an opening **108** and a surface of latch **40** defines an opening **110**, as shown in FIG. **9**. A pin **112** is configured for disposal within openings **108** and **110** to rotatably engage latch **40** with handle **68**. Engagement between handle **68** and latch **40** creates a pivot point **P5**, as shown in FIG. **6**. A biasing member, for example, a torsion spring **114** is configured for disposal with pin **112** and engagement to latch **40**, as shown in FIG. **9**. Spring **114** is configured to provide torque to latch **40** when pivoted into a non-locked and/or a locked orientation. An end **116** of spring **114** is configured for engagement to an indent **118** of latch **40** and an end **120** of spring **114** is configured for engagement to a pin **122**, as shown in FIG. **9**.

(54) Latch **40** includes an outer surface that defines a trigger **124** including a finger engagement surface, as shown in FIG. **2**. Trigger **124** is configured for engagement with a user such that latch **40** can be manually depressed to rotatably translate latch **40** into a selected orientation, as described herein. In some embodiments, the finger engagement surface can have one or more various surface configurations, for example, rough, threaded, arcuate, undulating, porous, semi-porous, dimpled, polished and/or textured.

(55) Latch **40** includes a surface that defines a slot **126** configured for engagement to a pin **128** disposed in a cavity **130** of shaft **38**. Latch **40** is rotatable relative to actuator **36** for capture of shaft **30** in the intermediate position such that pin **128** is disposed in slot **126** in a locked orientation, as shown in FIG. **6** and described herein. In some embodiments, slot **126** includes a C-groove

configuration. In some embodiments, slot **126** may have various cross-section configurations, for example, oval, oblong, triangular, rectangular, square, polygonal, irregular, uniform, non-uniform, variable, tubular and/or tapered. In some embodiments, slot **126** has one or more of various surface configurations, for example, rough, threaded, arcuate, undulating, porous, semi-porous, dimpled, polished and/or textured.

(56) Shaft **38** includes a surface that defines an opening **132**, as shown in FIG. **1** that is configured for movable disposal of latch **40** such that latch **40** can translate through shaft **38** to engage pin **128**. In some embodiments, opening **132** may have various cross-section configurations, for example, oval, oblong, triangular, rectangular, square, polygonal, irregular, uniform, non-uniform, variable, tubular and/or tapered.

(57) Latch **40** defines a ramp **134** and a ramp **136** disposed at a selected angular orientation α relative to ramp **134**, as shown in FIGS. **2**, **6**, **9** and **11**. Ramps **134**, **136** are configured for slidable engagement with pin **128** to facilitate movement of actuator **36** relative to shaft **30** between the positions, as described herein. Slidable engagement between ramps **134**, **136** and pin **128** is actuated via spring **114**. In some embodiments, ramps **134**, **136** enable actuator **36** to fully return to the open position from the closed position without actuator **36** becoming disposed in the intermediate position. Spring **62** provides the energy to return actuator **36** to the open position. In some embodiments, ramps **134**, **136** are configured to enable inserter **12** to automatically return to the open position from the closed position. In some embodiments, ramps **134**, **136** are configured to prevent inserter **12** from capture in the intermediate position via engagement between pin **128** and slot **126**, as described herein. Ramps **134**, **136** are relatively oriented to form angle α , as shown in FIG. **11**. In some embodiments, angle α is in a range from greater than 0 to 90 degrees.

(58) In the open position, as shown by arrows J in FIG. **1**, latch **40** is oriented with actuator **36** in the non-locked orientation where pin **128** engages an end surface **138** of latch **40**, as shown in FIG. **2**. In the intermediate position, as shown by arrows C in FIG. **5**, actuator **36** is manually compressed and slot **126** translates relative to pin **128** such that a surface of slot **126** engages pin **128** to orient latch **40** into the locked orientation, as shown in FIG. **6**. Latch **40** is depressed, as shown by arrow F in FIG. **9**, and spring **114** is biased and pin **128** engages ramp **136**, as shown in FIG. **9**. In the closed position, as shown by arrows G in FIG. **10**, actuator **36** is manually compressed and pin **128** translates from ramp **136** to ramp **134** such that latch **40** is oriented into the non-locked orientation, as shown in FIG. **11**.

(59) Ramps **134**, **136** enable inserter **12** to automatically return to an open position from the closed position and prevents inserter **12** from capture in the intermediate position. When actuator **36** is further compressed and released, inserter **12** automatically returns to the open position from the closed position. In some embodiments, actuator **36** is further compressed from the closed position and rapidly released to automatically return to the open position. In some embodiments, actuator **36** is further compressed from the closed position and slowly released such that actuator **36** returns to the intermediate position and latch **40** is depressible to return actuator **36** to the open position.

(60) Latch **40** includes a slot **138**, as shown in FIG. **6**. Slot **138** includes an end, for example, a rotatable limit **140** and an end, for example a rotatable limit **142**. Limits **140**, **142** are configured to limit the rotation of latch **40** relative to actuator **36**. Handle **68** includes an opening **144**, as shown in FIG. **5**. A pin **146** is configured for disposal within slot **138** and opening **144** such that when latch **40** rotates, pin **146** abuts limits **140**, **142**. When actuator **36** is in the open position and latch **40** is in the non-locked orientation, pin **146** abuts with limit **142**. In the intermediate position, latch **40** is disposed in the locked orientation such that pin **146** abuts limit **142**. In the non-locked orientation, latch **40** is depressed via trigger **124** and pin **146** is disposed within an intermediate section **148**, as shown in FIG. **9**. In the closed position, latch **40** is disposed in the non-locked orientation, and pin **146** abuts limit **140**, as shown in FIG. **11**.

(61) In operation, handles **68**, **70** are movable to the open position, as shown by arrows J in FIG. **1**, and latch **40** is disposed with actuator **36** in the non-locked orientation, as shown in FIG. **2**.

Receiver **14** is loaded into end **22** of inserter **12**. In the open position, tabs **42, 44** are in a biased radially inward direction, as shown by arrows B in FIG. **4**, and tabs **42, 44** do not engage groove **50** of receiver **14**.

(62) Handles **68, 70** of actuator **36** are movable to the intermediate position, as shown by arrows C in FIG. **5**, and latch **40** is disposed in the locked orientation, as shown in FIG. **6**. Latch **40** locks with pin **128** via slot **126** to prevent handles **68, 70** from opening or closing. In the intermediate position, receiver **14** is secured to end **22** of inserter **12**, as shown in FIG. **7**. Shaft **30** translates in a direction, for example, axially, as shown by arrow E in FIG. **7** and engages an inner surface of tabs **42, 44** of sleeve **26** to expand tabs **42, 44** to engage groove **50** of receiver **14**.

(63) Trigger **124** is depressed, as shown by arrow F in FIG. **9** and handles **68, 70** are compressed, for example, in an inward direction as shown by arrows G in FIG. **10**, such that shaft **30** translates axially, as shown by arrow H in FIG. **12** and sleeve **26** remains engaged to groove **50**. Handles **68, 70** are movable to the closed position, as shown by arrows G in FIG. **10**, and latch **40** is disposable in the non-locked orientation, as shown in FIG. **11**. End **54** of shaft **30** drives crown **64** axially, as shown by arrow I in FIG. **13**. Handles **68, 70** are further compressed from the closed position and rapidly released to automatically return to the open position, and receiver **14** is released from end **22**.

(64) In assembly, operation and use, surgical system **10**, is employed with a surgical procedure, for example, a surgical treatment of an applicable condition or injury of an affected section of a spinal column and adjacent areas within a body. In some embodiments, surgical system **10** can be used in any surgical method or technique including open surgery, mini-open surgery, minimally invasive surgery and percutaneous surgical implantation, whereby vertebrae V is accessed, such as through a mini-incision, and possibly also via a sleeve (not shown) that provides a protected passageway to vertebrae V. Once access to the surgical site is obtained, the particular surgical procedure can be performed for treating the spine disorder.

(65) An incision is made in the body of a patient and a cutting instrument (not shown) creates a surgical pathway. A preparation instrument (not shown) can be employed to prepare tissue surfaces of or surrounding vertebrae V, as well as for aspiration and irrigation of a surgical region. Pilot hole(s) (not shown) are made with the selected areas of bone, for example vertebrae for receiving shaft **18**.

(66) Handles **68, 70**, as described herein, are movable to the open position, as shown by arrows J in FIG. **1**, and latch **40** is disposed with actuator **36** in the non-locked orientation, as shown in FIG. **2**. Receiver **14** is loaded to end **22** of inserter **12**, as described herein, in a direction shown by arrow A in FIG. **3**. Handles **68, 70** of actuator **36** are movable to the intermediate position, as shown by arrows C in FIG. **5**, and latch **40** is disposed in the locked orientation, as shown in FIG. **6**. In the intermediate position, receiver **14** is secured to end **22** of inserter **12**, as shown in FIG. **7**. Shaft **30** translates axially, as shown by arrow E in FIG. **7** and engages an inner surface of tabs **42, 44** of sleeve **26** to expand tabs **42, 44** to engage groove **50** of receiver **14**.

(67) Handles **68, 70** are movable to the closed position, as shown by arrows G in FIG. **10**, and latch **40** is disposable in the non-locked orientation, as shown in FIG. **11**. Trigger **124** is depressed, as shown by arrow F in FIG. **9** and handles **68, 70** are compressed in an inward direction as shown by arrows G in FIG. **10**, such that shaft **30** translates axially, as shown by arrow H in FIG. **12** and sleeve **26** remains engaged to groove **50**. In the closed position, end **54** of shaft **30** drives crown **64** axially, as shown by arrow I in FIG. **13**. Handles **68, 70** are further compressed from the closed position and rapidly released to automatically return to the open position, and receiver **14** is released from end **22**.

(68) Upon completion of a procedure, inserter **12**, surgical instruments and/or tools, assemblies and non-implanted components of surgical system **10** are removed and the incision(s) are closed. One or more of the components of surgical system **10** can be made of radiolucent materials such as polymers. Radiomarkers may be included for identification under x-ray, fluoroscopy, CT or other

imaging technologies. In some embodiments, the use of surgical navigation, microsurgical and image guided technologies may be employed to access, view and repair spinal deterioration or damage, with the aid of surgical system **10**.

(69) In some embodiments, surgical system **10** may include one or a plurality of bone fixation devices, including plates, connectors, spinal rods and/or bone fasteners for use with a single vertebral level or a plurality of vertebral levels. In some embodiments, one or more of bone fixation devices may be engaged with tissue in various orientations, for example, series, parallel, offset, staggered and/or alternate vertebral levels. In some embodiments, one or more bone fixation devices may comprise multi-axial screws, sagittal angulation screws, pedicle screws, mono-axial screws, uni-planar screws, facet screws, fixed screws, tissue penetrating screws, conventional screws, expanding screws, wedges, anchors, buttons, clips, snaps, friction fittings, compressive fittings, expanding rivets, staples, nails, adhesives, posts, fixation plates and/or posts.

(70) In some embodiments, surgical system **10** includes an agent, which may be disposed, packed, coated or layered within, on or about the components and/or surfaces of surgical system **10**. In some embodiments, the agent may include bone growth promoting material, for example, bone graft to enhance fixation of the fixation elements with vertebrae. In some embodiments, the agent may be HA coating. In some embodiments, the agent may include one or a plurality of therapeutic agents and/or pharmacological agents for release, including sustained release, to treat, for example, pain, inflammation and degeneration.

(71) In some embodiments, as shown in FIG. **14**, surgical system **10** includes an implant support, for example, an extender **160** that includes a spinal rod reducer **162**. Extender **160** and reducer **162** are configured for use with inserter **12**. In some embodiments, extender **160** is oriented for manipulation, alignment and/or capture of receiver **14** and reducer **162** is configured to dispose a spinal rod (not shown) with receiver **14**. Reducer **162** includes an inner surface that defines a passageway **163** that is configured for disposal of sleeve **26** of inserter **12**. An outer surface **164** is threaded to an inner surface of extender **160** that defines a passageway **166**. Reducer **162** is rotated to translate reducer **162** axially, in a proximal or distal direction relative to inserter **12** and/or extender **160**. Reducer **162** is translated such that an end surface **168** engages the spinal rod in a configuration to move the spinal rod relative to receiver **14** to drive and/or reduce the spinal rod into receiver **14**.

(72) In some embodiments, as shown in FIGS. **15-22**, surgical system **10**, includes an inserter **212**, similar to inserter **12** described herein, configured for use with receiver **14**, as shown in FIG. **17**. Inserter **212** includes a proximal end **220** and a distal end **222**, as shown in FIG. **15**. Inserter **212** extends along and defines a longitudinal axis X2, as shown in FIG. **16**.

(73) Inserter **212** includes an outer sleeve **226** and an inner shaft **230**, as shown in FIG. **18**. Sleeve **226** and shaft **230** are configured for engagement to receiver **14**. Sleeve **226** includes an end **232** and an end **234**. In some embodiments, sleeve **226** may have various cross-section configurations, for example, oval, oblong, triangular, rectangular, square, polygonal, irregular, uniform, non-uniform, variable, tubular and/or tapered.

(74) End **232** is configured for engagement to an actuator **236**, similar to actuator **36**, as shown in FIG. **18**. Actuator **236** is movable relative to sleeve **226**, as described herein. An outer shaft **238** is disposed at end **232** and is configured for engagement to actuator **236** and a latch **240**, similar to latch **40** described herein, as shown in FIG. **16**. In some embodiments, shaft **238** may have various cross-section configurations, for example, oval, oblong, triangular, rectangular, square, polygonal, irregular, uniform, non-uniform, variable, tubular and/or tapered.

(75) End **234** is configured for engagement to an inner surface of receiver **14**, as shown in FIG. **17**. End **234** is expandable to engage the inner surface of receiver **14** to secure receiver **14** to inserter **12**, as shown in FIG. **17**. End **234** includes a spring tab **242** and a spring tab **244**. Tab **242** opposes tab **244**. Tabs **242** and **244** are flexible, biased radially inward and are expandable to engage the inner surface of receiver **14** and to secure receiver **14** to inserter **212**. Tab **242** includes an outer

surface that defines a projection **246** and tab **244** includes an outer surface that defines a projection **248**, as shown in FIG. **18**. Groove **50** of receiver **14** is configured for engagement to projections **246** and **248**, as shown in FIG. **18**. In some embodiments, sleeve **226** includes one or more spring tabs.

(76) End **234** includes an inner surface that defines a groove **245** and a groove **247**, as shown in FIG. **19**. Grooves **245**, **247** are configured for engagement to an end **254** of shaft **230** to facilitate engagement of crown **64**, as described herein.

(77) Shaft **230** includes an end **252** and end **254**, as shown in FIGS. **16** and **18**. Shaft **230** is in co-axial alignment relative to sleeve **226** and extends along longitudinal axis X2, as shown in FIG. **16**. Shaft **230** translates relative to sleeve **226** to secure receiver **14** to shaft **18**, as described herein. Shaft **230** is configured to expand sleeve **226** into engagement to the inner surface of receiver **14** to secure receiver **14**. In some embodiments, shaft **230** may have various cross-section configurations, for example, oval, oblong, triangular, rectangular, square, polygonal, irregular, uniform, non-uniform, variable, tubular and/or tapered.

(78) End **252** is configured for engagement to an end **256** of shaft **238**, as shown in FIG. **16**, similar to end **52** engagement to end **56** of shaft **38**, as described above with regard to inserter **12**. Shaft **230** is fixed to shaft **238** and shaft **230** is movable relative to sleeve **226**. A biasing member, for example, a spring **262** is configured for disposal about end **256**, as shown in FIG. **16** and is configured to provide energy in an axial direction to facilitate return movement of shaft **230** when actuator **236** is released, as described herein.

(79) End **254** is configured for engagement to crown **64** disposed within cavity **66** of receiver **14**, as shown in FIG. **18**. Crown **64** is configured for locking receiver **14** to shaft **18**, as described herein. In some embodiments, end **254** may have various cross-section configurations, for example, oval, oblong, triangular, rectangular, square, polygonal, irregular, uniform, non-uniform, variable, tubular and/or tapered.

(80) End **220** of inserter **212** includes actuator **236**, as shown in FIG. **15**. Actuator **236** is rotatable relative to sleeve **26** and is disposable between an open position including a non-locked orientation, an intermediate position including a locked orientation and a closed position including a non-locked orientation, as described herein. Actuator **236** includes a pair of lever handles **268**, **270**, as shown in FIG. **15** that are rotatable relative to sleeve **226**.

(81) Latch **240** is connected to actuator **236** in a non-locked orientation such that actuator **236** is movable relative to sleeve **226** in the open position and closed position, and a locked orientation such that actuator **236** is fixed relative to sleeve in the intermediate position. Latch **240** is connected to actuator **236** via handle **268**, in the same manner as latch **40** is connected to actuator **36**, described above with regard to FIGS. **1-13**.

(82) In operation, handles **268**, **270** are movable to the open position, and latch **240** is disposed with actuator **236** in the non-locked orientation. Receiver **14** is loaded into end **222** of inserter **212**. In the open position, tabs **242**, **244** are in a biased radially inward direction, as shown by arrows K in FIG. **18**, and tabs **242**, **244** do not engage groove **50** of receiver **14**.

(83) Handles **268**, **270** of actuator **236** are movable to the intermediate position, and latch **240** is disposed in the locked orientation. In the intermediate position, receiver **14** is secured to end **222** of inserter **212**, as shown in FIG. **21**. Shaft **230** translates in a direction, for example, axially, as shown by arrow M in FIG. **21** and end **254** engages grooves **245**, **247** and inner surfaces of tabs **242**, **244** of sleeve **226** to expand tabs **242**, **244** to engage groove **50** of receiver **14**.

(84) Handles **268**, **270** are movable to the closed position, and latch **240** is disposable in the non-locked orientation. A trigger **264** of latch **240** is depressed and handles **268**, **270** are compressed, for example, in an inward direction, such that shaft **230** translates axially, as shown by arrow N in FIG. **22** and sleeve **226** remains engaged groove **50**. End **254** of shaft **230** drives crown **64** axially, as shown by arrow N in FIG. **22**. Handles **268**, **270** are further compressed from the closed position and rapidly released to automatically return to the open position, and receiver **14** is released from

end 222.

(85) It will be understood that various modifications may be made to the embodiments disclosed herein. Therefore, the above description should not be construed as limiting, but merely as exemplification of the various embodiments. Those skilled in the art will envision other modifications within the scope and spirit of the claims appended hereto.

Claims

1. A surgical inserter comprising: an outer sleeve having a distal portion, the distal portion including at least one expandable projection being engageable with an inner groove of a spinal rod receiver of a bone fastener, the spinal rod receiver comprising a crown; an inner shaft having a distal portion being engageable with the distal portion of the outer sleeve to expand the at least one expandable projection into the inner groove to capture the spinal rod receiver; the inner shaft being engageable with the crown for connecting the spinal rod receiver with the surgical inserter; and the distal portion of the inner shaft including a tapered configuration and a tip engageable with the crown.
2. The surgical instrument of claim 1, wherein the distal portion of the inner shaft is translatable relative to the distal portion of the outer shaft.
3. The surgical instrument of claim 1, wherein the distal portion of the outer sleeve is biased radially inward.
4. The surgical instrument of claim 1, wherein the distal portion of the outer sleeve includes at least one spring tab biased radially inward and expandable to engage the inner groove and capture the spinal rod receiver.
5. The surgical instrument of claim 1, wherein the distal portion of the outer sleeve is engageable with the spinal rod receiver for snap fit assembly.
6. The surgical instrument of claim 1, wherein the at least one expandable projection includes a first expandable projection and a second expandable projection.
7. The surgical instrument of claim 1, wherein the distal portion of the outer sleeve includes opposing spring tabs biased radially inward and expandable to engage the inner groove and capture the spinal rod receiver.
8. A surgical inserter comprising: an outer sleeve having a distal portion, the distal portion including a first expandable projection and a second expandable projection being engageable with an inner groove of a spinal rod receiver of a bone fastener; and an inner shaft having a distal portion being engageable with the distal portion of the outer sleeve to expand the first expandable projection and the second expandable projection into the inner groove to capture the spinal rod receiver; wherein the spinal rod receiver comprises a crown and the distal portion of the inner shaft comprises an inwardly tapered configuration and a tip engageable with the crown, and wherein the distal portion of the outer sleeve is engageable with the spinal rod receiver for a snap fit assembly.
9. The surgical inserter of claim 8, wherein the first expandable projection and the second expandable projection are biased radially inward.
10. The surgical inserter of claim 8, wherein the distal portion of the outer sleeve includes opposing spring tabs biased radially inward and expandable to engage the inner groove and capture the spinal rod receiver.
11. The surgical inserter of claim 8, wherein the inner shaft is engageable with the crown for connecting the spinal rod receiver with the surgical instrument.
12. The surgical system of claim 8, wherein each of the first expandable projection and the second expandable projection are disposed in a spaced apart relation to the crown of the spinal rod receiver when the first expandable projection and the second expandable projection are engaged with the inner groove of the spinal rod receiver and when the inner shaft is engaged with the crown.
13. A surgical system comprising: a bone fastener shaft configured for fixation to vertebral tissue; a

spinal rod receiver configured for connection to the bone fastener shaft and comprising a crown disposed within a cavity of the spinal rod receiver; and a surgical instrument including an outer sleeve and an inner shaft, a distal portion of the outer sleeve including at least one expandable projection being engageable with an inner groove of the spinal rod receiver, and a distal portion of the inner shaft being engageable with the distal portion of the outer sleeve to expand the at least one expandable projection into the inner groove to capture the spinal rod receiver, wherein the distal portion of the inner shaft comprising a tapered configuration and a round tip for connecting the spinal rod receiver with the surgical instrument; wherein the distal portion of the inner shaft is disposed in an at least partially spaced apart relation to an interior of the outer sleeve and the round tip extends beyond the at least one expandable projection and is engageable with the concave surface of the crown when the at least one expandable projection is engaged with the inner groove of the spinal rod receiver.

14. The surgical system of claim 13, wherein the distal portion of the inner shaft is translatable relative to the distal portion of the outer-shaft sleeve.

15. The surgical system of claim 13, wherein the distal portion of the outer sleeve includes at least one spring tab biased radially inward and expandable to engage the inner groove and capture the spinal rod receiver.

16. The surgical system of claim 13, wherein the distal portion of the outer sleeve includes opposing spring tabs biased radially inward and expandable to engage the inner groove and capture the spinal rod receiver.

17. The surgical system of claim 13, wherein the inner shaft is disposed in an at least partially spaced apart relation to an interior of the at least one expandable projection when the at least one expandable projection is engaged with the inner groove of the spinal rod receiver.

18. The surgical system of claim 13, wherein the at least one expandable projection comprises two pairs of oppositely disposed expandable projections.

19. The surgical system of claim 13, wherein the at least one expandable projection comprises four expandable projections; each expandable projection being perpendicularly disposed to an adjacent expandable projection.

20. The surgical system of claim 13, wherein the distal portion of the inner shaft is disposed in an at least partially spaced apart relation to the at least one expandable projection when the at least one expandable projection is engaged with the inner groove of the spinal rod receiver.
